

Performance Metrics

It is time to completely rethink the way we measure home and building efficiency. We hear a lot these days about buildings being certified as “green” or about major infrastructure changes intended to make them “efficient.” But truly conserving energy is a matter of results—it means analyzing the daily performance metrics, tweaking the building functions as needed, and rewarding those individuals who are actually driving the building to peak operating efficiency.

Whether we are talking about a single-family house, apartments in multifamily buildings, or a large corporate facility, in each case a particular individual (or team) is making decisions about the HVAC system. It might be a homeowner with limited knowledge, an HVAC contractor, a building manager, or a facilities director. In each case, it is important for that person or that team to see himself, or themselves, as active drivers steering the home or building

daily toward specific goals and seeking to reach defined key performance metrics. This is not a passive activity in which you can fall asleep at the wheel. Whoever does it must be vigilant in tweaking the system daily toward peak operating efficiency.

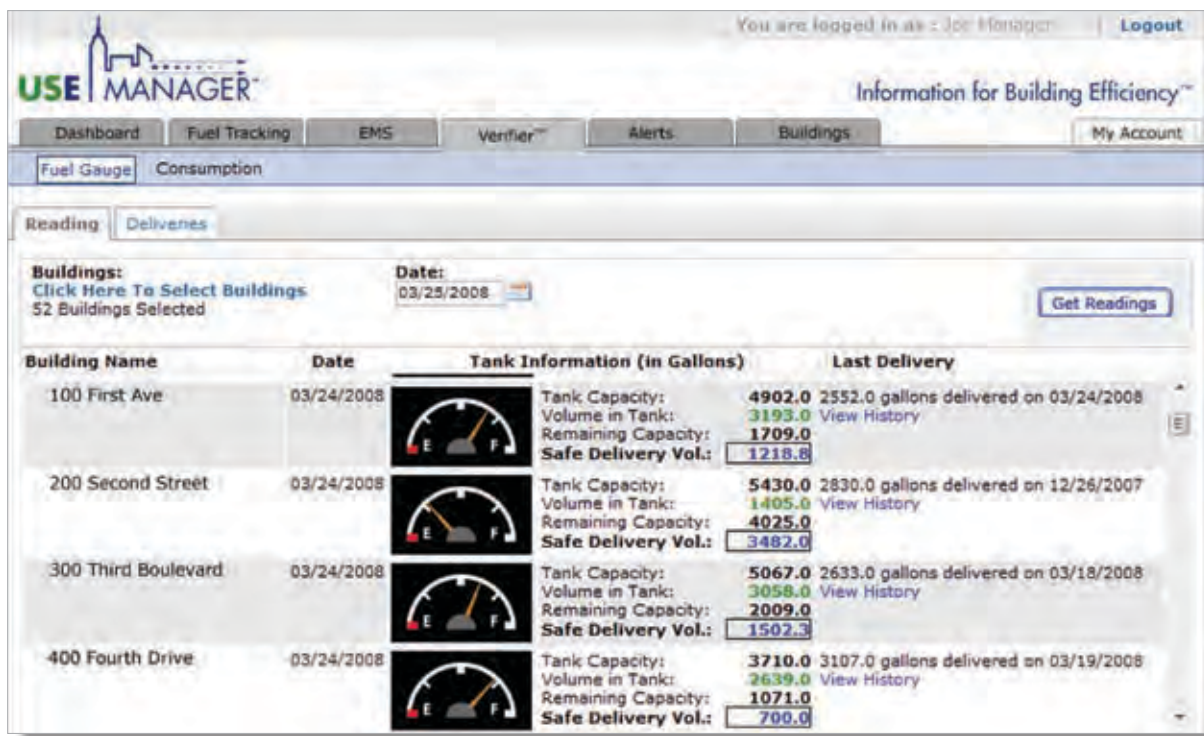
For years, single-family homes have used thermostats that control the heating system based on an analysis that includes the indoor temperature. This is important, because in order to conserve energy, you must control boiler run time. That said, most centrally heated multifamily buildings are still using an older system that only takes into account outdoor air temperature. For example, if it is 9 am and 25°F outside, the boiler is on. If it is 2:30 pm and 50°F outside, the boiler is still on; and by now, it is about 80°F inside and tenants have opened their windows. Open windows waste money and increase carbon emissions.

Controlling boiler run time is the single most effective way to retrofit a centrally heated building to be energy efficient. Today’s energy management systems, often referred to as EMS, can regulate boiler run time based on an

advanced algorithm that accounts for both outdoor and indoor temperatures. The system schedules the boiler to turn off and on as needed, regulating the temperature so that the building is not overheated.

While the first step is controlling boiler run time, the second step is monitoring the system effectively. This holds true for both single-family homes and multifamily buildings. It is crucial to monitor daily performance metrics from a results-oriented perspective and tweak the heating system functions as needed. For larger buildings, it helps to use a building-monitoring system that facilitates this daily oversight and provides alerts when the system is not running optimally.

Advanced online monitoring systems, such as U.S. Energy Group’s USE Manager, will provide alerts if the boiler run time is too long, or if the makeup water indicates a major leak. They also monitor mixing valve temperatures to make sure hot water isn’t scalding hot and track stack temperatures to ensure that heat is being transferred efficiently to the



US Energy’s USE Manager monitors heating systems, verifies oil deliveries, tracks fuel-use, and drives fuel savings for an entire portfolio of buildings.



U.S. Energy's Verifier Digital Fuel Gauge is an independent and accurate way to confirm a heating oil delivery, prevent theft, and manage your inventory and budget.

boiler and is not escaping through the chimney. Systems such as these are important for HVAC contractors who need to know exactly what is going on in several buildings at one time, because they immediately draw one's attention to specific concerns or violations and provide the steps to assign repairs and maintenance. Monitoring systems track key performance data and make it easier to analyze how the entire heating system is working.

Jordan Platt of Kaled Management in New York set himself the particular goal of saving money on oil. By using an energy management system, the USE Controller EMS, and by actively driving his buildings, he was able to reduce boiler run time and prevent leaks and system malfunctions. He cut the number of deliveries of oil, and he estimates that he saved \$12,000–\$15,000 per building on a \$10,000 system investment.

David Diamond, president and CEO of Rosenberg-Diamond Development, owns 50 buildings in the Bronx and has been actively driving his buildings for years. In the beginning, through daily monitoring, he saved about 40% on fuel consumption. Now, he has six staff members who actively address alerts and he continues to realize approximately 30% savings every year. He is so vigilant about making sure his HVAC systems are running optimally that he's been known to use a laptop to tweak them while on vacation. Now that's an active driver!

Using performance metrics is about results—saving \$15,000 is a concrete result and reducing fuel consumption by 40% is a concrete result. The active driver can set specific goals and monitor the heating system daily to achieve provable results. We are so busy certifying buildings as energy efficient or labeling homes as “green”—but it is crucial that we evaluate true conservation measures that can be analyzed through concrete performance metrics and acknowledge the efficiency drivers who are behind the wheel. **H_e**

—Thomas Scali

Thomas Scali is the director of sales and business development for U.S. Energy Group.

For more information:

To learn more about the U.S. Energy Group's USE Manager, go to www.use-group.com.